

Aadarsha Gopala Reddy

+1 (740) 802-1776 | adurs2002@gmail.com | linkedin.com/in/agopalareddy | github.com/agopalareddy | agreddy.com

EDUCATION

Washington University in St. Louis <i>M.S. Computer Science</i> Thesis: Gopala Reddy, Aadarsha, “Toward Vehicle-Agnostic Driving Signatures for Cognitive Impairment Prediction from Naturalistic Driving Data” (2026). <i>McKelvey School of Engineering Graduate Student Theses & Dissertations</i>. 1341. https://doi.org/10.7936/g95n-fv35	St. Louis, Missouri, USA August 2024 - May 2026
Ohio Wesleyan University <i>B.A. Computer Science & Data Analytics</i> Minor: Economics	Delaware, Ohio, USA August 2020 - May 2023

RESEARCH EXPERIENCE

Washington University in St. Louis <i>Toward Vehicle-Agnostic Driving Signatures for Cognitive Impairment Prediction from Naturalistic Driving Data</i> DRIVES Project (PI: Dr. Ganesh Babulal; Advisors: Dr. Alvitta Ottley, Dr. Nathan Jacobs) - Built an end-to-end pipeline processing 26,968 participant-weeks of naturalistic driving data from 304 participants, deriving vehicle-agnostic weekly behavioral features and integrating demographic covariates for CDR status prediction. - Compared six CDR prediction models (GRU-DANN, DANN, logistic regression, random forest, XGBoost, MLP) under leave-one-participant-out cross-validation; GRU-DANN achieved highest participant-level ROC AUC (0.599) and balanced accuracy (0.584).	St. Louis, Missouri, USA August 2025 - April 2026
Ohio Wesleyan University <i>Opinion Survey on Artificial Intelligence in the Workplace</i> Independent Study (Mentor: Dr. Nicholas Dietrich) - Designed, fielded, and investigated a survey experiment on the impact of AI on 250 employees within each of four industries using survey data. - Analyzed data using Tableau dashboards and R to identify trends and common opinions. - Presented preliminary results on AI’s positive impact on employee work at the OWU Spring Student Symposium. <i>Artificial Intelligence in Modern Board Games — Lost Cities AI</i> Summer Science Research Program (Mentor: Dr. Sean McCulloch) - Developed a digital version of the Lost Cities card game using Java. - Implemented an intelligent agent that played the game against a human, winning 13 of 18 games in testing.	Delaware, Ohio, USA January 2023 - May 2023 May 2022 - July 2022

TEACHING & MENTORSHIP EXPERIENCE

Washington University in St. Louis <i>Graduate Teaching Assistant - CSE 5114: Data Manipulation and Management at Scale</i> - Taught large-scale data pipeline concepts (Snowflake, Airflow, Spark, Kafka, Flink) to a graduate cohort, covering fault-tolerant design, streaming windowing strategies, and query optimization. - Designed oral midterms as mock technical interviews, requiring students to reason through pipeline architecture decisions and troubleshoot system failures under real-world constraints. - Mentored three capstone teams from raw open datasets to production-grade dashboards, providing weekly feedback that kept all teams on schedule through final presentations. - Held weekly office hours to debug student pipeline implementations across Snowflake schema design, Airflow DAG failures, and Kafka consumer configuration. <i>HackWashU Databases Workshop</i> - Designed and facilitated a hands-on databases workshop for beginner-to-intermediate developers, teaching through two complete projects rather than lectures. - Guided participants through building a normalized SQLite database from the iTunes Search API and a full-stack Supabase Todo app with Row Level Security — covering relational databases, APIs, cloud Postgres, and access control. <i>Engineering Workshop by AGES & Science Coach</i> - Delivered an introductory workshop on artificial intelligence to high school students, covering core AI concepts with hands-on demonstrations.	St. Louis, Missouri, USA January 2026 - May 2026 September 2025 September 2024
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Vice President of Finance, Campus Programming Board

January 2022 - December 2022

- Managed annual budgets exceeding \$86,000 for campus-wide programming and secured \$10,000 in additional annual funding through negotiation with WCSA.

PROJECTS

Washington University in St. Louis

St. Louis, Missouri, USA

Red-Blue Visual Auto Defender: Automated Visual Jailbreak Generation and Explainable Defenses

August 2025 - December 2025

- Built an automated red-blue teaming pipeline for vision-language model (VLM) security, addressing visual prompt injection (VPI) attacks against VLM-based agents.
- Generated visual jailbreak attack images by overlaying malicious instructions onto benign images and evaluated them against a target VLM (Gemma-3-4b-it) using semantic response analysis.
- Auto-generated deterministic, OCR-based Python defenses for each successful attack, validated with accuracy, precision, and recall metrics in an iterative refinement loop.

Multimodal Prediction of Alzheimer’s Disease

August 2024 - December 2024

- Developed a multimodal fusion approach for early detection of Alzheimer’s Disease, combining MRI imaging and clinical/demographic data from the OASIS-1 dataset.
- Implemented CNNs using TensorFlow/Keras for brain MRI analysis alongside XGBoost and Random Forest models for tabular clinical data.
- Created a combined classifier integrating both modalities and documented the project in NeurIPS format with comprehensive performance evaluation metrics and visualizations.

Datacenter Cooling Optimization using Deep Reinforcement Learning

August 2024 - December 2024

- Implemented multiple deep reinforcement learning algorithms (DDQN, PPO, SAC) integrated with EnergyPlus simulations to optimize datacenter cooling systems.
- Achieved 35.8% energy efficiency improvement over baseline using DDQN, addressing the underserved small to mid-sized datacenter market.
- Created a comprehensive framework integrating building energy simulation with reinforcement learning for dynamic thermal management.

MITxSureStart FutureMakers Create-a-Thon Program

Remote

Parkinson’s Disease AI Diagnosis Software

June 2022 - July 2022

- Built a dual-input pipeline combining audio spectrograms (custom CNN for vocal tremor detection) and spiral-drawing images (Random Forest classifier with augmentation for micrographia detection).
- Implemented a weighted ensemble combining both modalities into a single Parkinson’s diagnosis score.
- Designed a Flask web interface for user submissions of audio and drawing samples for real-time risk assessment.

OTHER PROFESSIONAL EXPERIENCE

Crittero, Inc.

Remote

AI Engineer Intern

June 2025 - August 2025

- Generated 100,000 labeled social media posts across seven user personas via time-series behavioral modeling in TypeScript, capturing post sequences, engagement timings, and content preferences.
- Trained a TensorFlow/Keras recommendation model achieving 80% accuracy (30-percentage-point gain over random baseline) with persona-type, content-category, and temporal embeddings, plus a collaborative-filtering cold-start fallback for new users.
- Built a configurable Python CLI test harness (700+ lines) with deterministic seeding and CSV export for reproducible model evaluation, using it to validate accuracy across manually tuned embedding-dimension and cold-start-threshold configurations.

Washington University in St. Louis

St. Louis, Missouri, USA

Graduate Assistant, Taylor Family Center for Student Success

August 2024 - May 2026

- Planned and ran three recurring engagement programs for first-generation, limited-income (FLI) students — Welcome Dinner, First-Gen Week, and End of Week Unwind — coordinating logistics, promotion, and on-the-day operations.
- Authored FERPA-compliant process manuals for engagement data collection and evaluation, giving center staff a standardized, repeatable framework for measuring FLI student participation.
- Analyzed semester survey responses and event attendance data, producing reports that directly informed the following cycle’s programming decisions and outreach priorities.
- Managed the center’s weekly newsletter; represented the department at off-campus excursions and FLI student trips.

- Analyzed IoT environmental sensor data streams on AWS, building dashboards that surfaced chemical imbalances and equipment anomalies through automated threshold detection and trend analysis.
- Built AWS data ingestion pipelines to normalize and store continuous sensor telemetry from deployed devices, supporting real-time operational monitoring of deployed IoT fleets.
- Wrote SQL queries to aggregate historical sensor data by pool, date range, and chemical parameter, powering the trend analytics and anomaly detection shown in client dashboards.
- Designed and engineered the Figma-to-React UI for an all-in-one environmental sensor device, delivering a pixel-perfect, production-ready interface from initial wireframes.

TECHNICAL SKILLS & LANGUAGES

Programming Languages: Python, TypeScript, JavaScript, SQL, Java, C++, R, C#, Rust.

Frameworks & Libraries: PyTorch, TensorFlow, Scikit-learn, Pandas, NumPy, Matplotlib, Seaborn, React, Vue.js, Node.js, Express.js, FastAPI, Socket.IO, Streamlit.

Tools & Technologies: L^AT_EX, Git, Tableau, Power BI, AWS, Docker, PostgreSQL, MySQL, MongoDB, Snowflake, Apache Airflow, Spark, Kafka, Flink, Gemini/OpenAI APIs, Microsoft 365, Google Workspace.

Languages: Proficient in English, Kannada, Telugu; Conversational in Hindi.